OPHIA
UMHACKATHON

AI-Powered Clinical Workflow Intelligence

Transforming Malaysia's fragmented hospital workflows into an intelligent, stateful care system powered by GROCQ.

UMHACKATHON 2026 · DOMAIN 1: AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

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THE PROBLEM

Hospital fragmentation in Malaysia

Fragmented Records

Patient data is scattered across paper notes, standalone EMRs, and siloed departments - no unified view.

73%

of Malaysian public hospitals still rely on paper-based patient records

Unstructured Doctor Notes

Handwritten notes and voice memos are never digitised. Critical data is lost or misread.

3X

longer average discharge time vs. digitised hospitals

Manual Decision Chains

Bed reservations, follow-ups, and discharge summaries require manual coordination - causing dangerous delays.

68%

of adverse events linked to communication breakdowns

INTRODUCING

OPHIA

- **READ**
  HANDWRITING + VOICE → STRUCTURED TEXT VIA GLM-OCR
- **REASON**
  GLM-4 EXTRACTS ENTITIES, FLAGS RISKS, DECIDES NEXT ACTIONS
- **ACT**
  APIS TRIGGER BED BOOKING, APPOINTMENTS, DISCHARGE PDF

A stateful, multi-agent clinical workflow engine built on Z.AI GLM

Ophia is not a chatbot.

It is an intelligent orchestration layer that:

→ Reads messy handwritten notes & voice memos via GLM-OCR

→ Reasons about patient state using grocq

→ Acts by triggering real hospital workflows — bed reservation, follow-ups, discharge summaries

Every step feeds the next. Remove GLM and the system cannot function.

5 STEPS

1. Doctor Inputs Notes — Photo, voice, or free text
2. GLM-OCR Extracts Text — Handwriting & audio → clean text
3. GLM-4 Structures Data — SOAP format, ICD-10 codes
4. GLM-4 Decides Action — Admit / discharge / follow-up
5. APIs Are Triggered — Bed, appointment & discharge PDF

If grocq is removed, steps 2–4 completely fail — GLM is the critical path, not a feature.

3 edge case chips:

1. Missing dosage? GLM asks doctor.
2. API failure? Fallback queued.
3. Ambiguous symptoms? GLM flags for review.

HOW IT WORKS

A SINGLE PATIENT VISIT TRIGGERS A FULLY AUTOMATED MULTI-STEP CHAIN

SYSTEM ARCHITECTURE

FOUR-LAYER STACK - EVERY LAYER POWERED OR ORCHESTRATED BYGROCQ

4 LAYERS:

- Frontend → React Dashboard - Doctor inputs notes · Views structured outputs · Receives safety alerts
- Ingestion → GLM-OCR + Whisper - Handwritten image & audio → clean raw text string
- AI Engine →GROCQ - Read-Reason-Act loop · Structures data · Routes workflow decisions
- Data & APIs → Supabase + Flask - 5 REST endpoints · Patient DB · Bed/appointment/discharge mock APIs

GLM CRITICAL PATH: OCR LAYER READS IT → GLM-4 REASONS ABOUT IT → MCP TOOLS ACT ON IT.
THREE HARD DEPENDENCIES.

MEDICAL-GRADE SAFETY SUBHEADER

GLM HANDLES AMBIGUITY, INCOMPLETE DATA, AND API FAILURES GRACEFULLY

- **Drug Allergy Block** — GLM cross-checks new prescriptions against patient allergy records in Supabase. Dangerous orders are blocked before they reach any nurse.
- **Incomplete Notes** — If GLM detects >2 critical missing fields, it returns a clarification request to the doctor — never guesses.
- **API Failure Fallback** — If bed-reservation API times out, GLM queues the action with a retry flag and notifies the nurse station.
- **Ambiguous Symptoms** — When diagnosis confidence is below threshold, GLM flags for senior physician review instead of auto-routing.
- **Stateful Context** — GLM remembers the full patient session — prior visits, current meds, allergies — across all 5 workflow steps.
- **Audit Trail** — Every GLM decision logged to Supabase with timestamp + reasoning summary, meeting Malaysian MOH requirements.

LIVE DEMO

BLOCKING A DANGEROUS PRESCRIPTION

PATIENT: AHMAD, 52 — PENICILLIN ALLERGY ON FILE

1. **Doctor** — Uploads handwritten note prescribing amoxicillin (a penicillin-class antibiotic)
2. **GLM-OCR** — Reads handwriting → extracts "amoxicillin 500mg TDS" as structured text
3. **GLM-4** — Queries Supabase patient record → finds penicillin allergy flag
4. **GLM-4** — Decision: BLOCK prescription · Generate safety alert · Notify doctor dashboard
5. **Dashboard** — Doctor receives red alert: "Amoxicillin contraindicated — penicillin allergy" with alternative suggestion
6. **GLM-4** — Logs incident to audit trail with timestamp + reasoning for MOH compliance

ROADMAP

FROM HACKATHON PROTOTYPE TO NATIONAL CLINICAL INFRASTRUCTURE

Phase 1 — *Hackathon MVP*

→ Doctor dashboard with 3 input modes
→ GLM-OCR handwriting + audio pipeline
→ GLM-4 Read-Reason-Act orchestration
→ 5 mock hospital APIs via Flask
→ Drug allergy safety block

Phase 2 — Pilot

→ Deploy in 1 Malaysian public hospital
→ Connect to real HIS (Hospital Info System)
→ Malay language support via GLM-4
→ MOH compliance audit logging
→ Physician feedback loop & retraining

Phase 3 — Scale

→ Roll out to KKM network (145 hospitals)
→ Multi-specialty workflow templates
→ Predictive bed management
→ Integration with MySejahtera health records
→ ASEAN regional expansion

Malaysia's hospitals deserve better than fragmented workflows.
Ophia puts grocq at the center of every clinical decision -
so doctors can focus on patients, not paperwork.

OPHIA

READ. REASON. ACT.